The Fractional Riccati–Müntz–Padé Theorem

Closed-Form Density and European Call Price under Rough Heston via Fox H, with COS Pricing as Alternative

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Abstract

Three results are established for the rough Heston model. First (Theorem 14), the diagonal $[M\,/\,M]$ Padé resummation of the local Puiseux series of the cumulant generating function converges on every compact subset of the complement of the Stahl branch/polar set, including beyond the Puiseux radius of the original series, at Stahl rate. The Padé denominator is the orthogonal polynomial of degree M under the Hankel moment functional, produced directly from the Müntz moments by the Chebyshev algorithm in arbitrary-precision arithmetic, equivalent to solving the Hankel system but without matrix formation or inversion. Second (Theorems 25, 27), both the density of the log-return and the European call price admit strictly closed-form representations as finite sums of Fox H-functions in the partial-fraction data of the rational characteristic exponent Φ_M, evaluable in $O(M \log^2 M)$ operations per strike at fixed maturity, with the front-end Chebyshev construction at $O(M^2)$ per parameter evaluation and the partial-fraction expansion at $O(M^2)$ per maturity. The price formula is derived by two analytically independent routes (convolution chain and Mellin–Barnes inversion) converging on identical parameter data. The Padé characteristic function has Gaussian real-axis decay, sharper than the stretched-exponential decay of the exact rough Heston cf. Third (Theorem 31), the same prices may alternatively be computed by applying the Fang–Oosterlee COS method to $\Phi_M(\cdot, T)$, with Gaussian-rate truncation error.

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1 Fractional calculus and the Müntz basis

Definition 1. *[Riemann–Liouville operators] For $\mu \in (0,1]$ and f locally integrable on $(0,\infty)$,*

$$I^\mu f(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s)\, d\, s, \tag{1}$$

$$D^\mu f(t) := \frac{d}{d\,t} (I^{1-\mu} f)(t). \tag{2}$$

Lemma 2. *[Power rule] For $s > -1$ and $\alpha > 0$,*

$$I^\alpha t^s = \frac{\Gamma(s+1)}{\Gamma(s+\alpha+1)}\, t^{s+\alpha}. \tag{3}$$

Proof. Substitution $\sigma = t\,\tau$ in I^α gives $I^\alpha t^s = t^{s+\alpha} B(\alpha, s+1) / \Gamma(\alpha)$, and $B(\alpha, s+1) = \Gamma(\alpha)\,\Gamma(s+1) / \Gamma(s+\alpha+1)$ gives (3). □

The Müntz lattice $\{t^{k\mu}\}_{k\geq 0}$ is closed under I^μ by (3) and closed under multiplication. In the variable $z = t^\mu$ every element is z^k, so ordinary power-series arithmetic applies.

2 The Müntz–Tau recurrence

Define $\gamma_k := \Gamma((k-1)\,\mu+1) / \Gamma(k\,\mu+1)$ for $k \geq 1$, so that $I^\mu t^{(k-1)\mu} = \gamma_k\, t^{k\mu}$.

Theorem 3. *[Müntz–Tau recurrence] The unique formal power-series solution of*

$$y = I^{\mu}[c_0 + c_1\, y + c_2\, y^2], \qquad y(0) = 0, \tag{4}$$

is $y(t) = \sum_{k \geq 1} a_k\, t^{k\mu}$ *with*

$$a_1 = \frac{c_0}{\Gamma(\mu+1)}, \tag{5}$$

$$a_k = \gamma_k \left[c_1\ a_{k-1} + c_2 \sum_{j=1}^{k-2} a_j\ a_{k-1-j} \right], \quad k \geq 2. \tag{6}$$

Proof. Substitute the ansatz into (4) and apply (3) termwise. Uniqueness by induction in k. □

3 Rough Heston cumulant moments

The rough Heston model has parameters $\theta, \lambda, \rho, \nu, V_0, H \in \mathbb{R}$ with $H \in (0, \frac{1}{2})$ and $\mu := H + \frac{1}{2} \in (\frac{1}{2}, 1)$. The log-return characteristic function is $\phi_T(v) := \mathbb{E}[e^{ivX_T}] = \exp(\Phi(v,T))$ with $X_T := \log(S_T / S_0) - r\,T$ and

$$\Phi(v,T) = \theta\,\lambda \int_0^T h(v,s)\ ds + V_0\, I^{1-\mu} h(v,\cdot)(T), \tag{7}$$

$h(v,\cdot)$ solving

$$h(v,t) = I^{\mu}[P(v) + Q(v)\, h(v,t) + R\ h(v,t)^2], \tag{8}$$

with $P(v) = \frac{1}{2}(-v^2 - i\,v)$, $Q(v) = \lambda\,(i\,v\,\rho\,\nu - 1)$, $R = \frac{1}{2}\lambda^2\,\nu^2$.

Lemma 4. *[Riccati Puiseux series] The unique formal solution of* (8) *is* $h(v,t) = \sum_{k \geq 1} a_k(v)\, t^{k\mu}$ *with* $\{a_k(v)\}$ *from Theorem 3 applied to* $c_0 = P(v), c_1 = Q(v), c_2 = R$. *Each* $a_k(v)$ *is a polynomial in* v *of degree at most* $k+1$ *satisfying* $a_k(0) = a_k(-i) = 0$.

Theorem 5. *[Cumulant generating function as Puiseux series] With* $z := T^{\mu}$, $\Phi(v,T) = T \cdot \hat{\Phi}(v,z)$,

$$\hat{\Phi}(v,z) = \sum_{k=0}^{\infty} M_k(v)\ z^k, \tag{9}$$

$$M_k(v) = \frac{\theta\,\lambda}{k\,\mu+1}\ a_k(v) + V_0\,\frac{\Gamma((k+1)\,\mu+1)}{\Gamma(k\,\mu+2)}\ a_{k+1}(v), \quad k \geq 0, \tag{10}$$

with $a_0(v) \equiv 0$.

Remark 6. [Index bookkeeping] Computing $\{M_k(v)\}_{k=0}^{2M-1}$ via (10) requires $\{a_k(v)\}_{k=1}^{2M}$, since the V_0 term in M_k uses a_{k+1}. The Müntz-coefficient budget is $2\,M$ Riccati coefficients per strike.

4 Padé resummation via the Chebyshev algorithm

The diagonal $[M/M]$ Padé approximant of $\hat{\Phi}$ is the unique pair (P_M, Q_M) with $\deg P_M \leq M-1$, $\deg Q_M \leq M$, $Q_M(0)=1$, satisfying

$$Q_M(z)\hat{\Phi}(v,z) - P_M(z) = O(z^{2M}). \tag{11}$$

4.1 Moment functional

Define $\mathcal{L}\colon \mathbb{C}[w] \to \mathbb{C}$ by $\mathcal{L}[w^k] := M_k(v)$, extended linearly. Let $H_n := \det\,(M_{i+j})_{0\leq i,j<n}$. Quasi-definiteness $H_n \neq 0$ $(n \geq 1)$ is assumed throughout.

4.2 Three-term recurrence and orthogonality

Theorem 7. *[Three-term recurrence] Under quasi-definiteness there is a unique sequence of monic polynomials* $\{p_n\}_{n\geq 0}$, $\deg p_n = n$, *satisfying*

$$\mathcal{L}\,[w^k\ p_n(w)] = h_n\ \delta_{k,n} \quad (0 \leq k \leq n), \quad h_n := \mathcal{L}\,[w^n\,p_n] \neq 0, \tag{12}$$

$$p_{n+1}(w) = (w-\alpha_n)\,p_n(w) - \beta_n\,p_{n-1}(w), \quad p_0 \equiv 1,\ p_{-1} \equiv 0, \tag{13}$$

with $\alpha_n = \mathcal{L}\,[w\,p_n^2]/h_n$, $\beta_n = h_n/h_{n-1}$ $(n \geq 1)$, $\beta_0 := 0$, $h_n = H_{n+1}/H_n$.

4.3 The Chebyshev algorithm

Theorem 8. *[Chebyshev algorithm]* $\sigma_{k,l} := \mathcal{L}\,[p_k(w)\,w^l]$ $(k \geq -1,\ l \geq 0)$.

$$\sigma_{-1,l} = 0, \qquad \sigma_{0,l} = M_l(v), \tag{14}$$

$$\sigma_{k+1,l} = \sigma_{k,l+1} - \alpha_k\,\sigma_{k,l} - \beta_k\,\sigma_{k-1,l}, \quad k \geq 0,\ l \geq k+1, \tag{15}$$

$$\alpha_0 = M_1(v)/M_0(v), \quad \beta_0 = 0, \quad h_0 = M_0(v), \tag{16}$$

$$\alpha_k = \frac{\sigma_{k,k+1}}{\sigma_{k,k}} - \frac{\sigma_{k-1,k}}{\sigma_{k-1,k-1}},\ \ \beta_k = \frac{\sigma_{k,k}}{\sigma_{k-1,k-1}},\ \ h_k = \sigma_{k,k}, \quad k \geq 1. \tag{17}$$

The cost is $O(M^2)$ *ring operations on* $\{M_l(v)\}_{l=0}^{2M-1}$*, with no Hankel matrix formed and no linear system solved. Exact in arbitrary precision.*

4.4 Padé numerator from associated polynomials

Definition 9. *[Associated polynomial of the second kind]*

$$p_n^{(1)}(w) := \mathcal{L}_v\!\left[\frac{p_n(w) - p_n(v)}{w - v}\right]. \tag{18}$$

Lemma 10. *[Same recurrence]* $p_{n+1}^{(1)} = (w - \alpha_n)\, p_n^{(1)} - \beta_n\, p_{n-1}^{(1)}$ *(*$n \geq 1$*),* $p_0^{(1)} \equiv 0$*,* $p_1^{(1)} \equiv M_0(v)$*.*

4.5 Padé matching

Lemma 11. *[Padé denominator and numerator]* $Q_M(z) := z^M\, p_M\,(1/z)$*,* $P_M(z) := z^{M-1}\, p_M^{(1)}\,(1/z)$*. Then* $Q_M(0) = 1$*,* $\deg P_M \leq M - 1$*, and*

$$Q_M(z)\hat{\Phi}(v, z) - P_M(z) = h_M\, z^{2M} + O(z^{2M+1}). \tag{19}$$

Corollary 12. *[Rational characteristic exponent]*

$$\Phi_M(v, T) := T \cdot z \cdot \frac{P_M(z; v, T)}{Q_M(z; v, T)}|_{z = T^\mu} \tag{20}$$

is rational in v *at fixed* T*, with* $2\,M$ *poles in* v *from* $Q_M(z; v, T)|_{z = T^\mu} = 0$*.*

4.6 J-fraction

Theorem 13. *[J-fraction]*

$$\hat{\Phi}(v, z) = \cfrac{M_0(v)}{1 - \alpha_0\, z - \cfrac{\beta_1\, z^2}{1 - \alpha_1\, z - \cfrac{\beta_2\, z^2}{1 - \alpha_2\, z - \cdots}}} \tag{21}$$

with M*-th convergent* $P_M(z)/Q_M(z)$*.*

5 Convergence and a-posteriori bound

Theorem 14. *[Stahl convergence] Let* $\rho(v) > 0$ *be the Puiseux radius and* $K(v) \subset \mathbb{C} \setminus [0, \rho(v))$ *the Stahl branch/polar set. Then*

$$|\hat{\Phi}(v, z) - R_M\,(z; v)|^{1/(2M)} \to e^{-\mathtt{g}(z, K(v))} \tag{22}$$

in capacity on $\mathbb{C} \setminus K(v)$*, uniform on real compacta. In particular* $\Phi_M(v, T) \to \Phi(v, T)$ *for every* $T \geq 0$ *with* $T^\mu \notin K(v)$*, including* T *beyond the Puiseux radius.*

Theorem 15. *[A-posteriori bound, supremum-ratio] Let* $\Delta_M := \Phi_M - \Phi_{M-1}$ *and* $q_M := \sup_{n \geq M} |\Delta_{n+1}|/|\Delta_n|$*, with* $q_M \to e^{-2\mathtt{g}(z, K(v))} < 1$ *on* $\mathbb{C} \setminus K(v)$*. Whenever* $q_M < 1$*,*

$$|\Phi(v, T) - \Phi_M(v, T)| \leq \frac{|\Delta_M|}{1 - q_M}. \tag{23}$$

Remark 16. [Aitken form] Under exact geometric decay $|\Delta_{M+1}|/|\Delta_M| = q$ the sharper Aitken form $|\Phi - \Phi_M| \le |\Delta_M|^2/(|\Delta_{M-1}| - |\Delta_M|)$ applies; in general it is heuristic.

Lemma 17. *[Normalisation]* $\Phi_M(0,T) = \Phi_M(-i,T) = 0$, *hence* $\phi_M(0,T) = \phi_M(-i,T) = 1$.

Lemma 18. *[Hermitian symmetry]* $\phi_M(-\bar{v},T) = \overline{\phi_M(v,T)}$.

Lemma 19. *[Gaussian real-axis decay] With* $(\sigma_T, \mu_T, \{u_j(T), A_j(T)\}_{j=1}^{2M})$ *from PFE* (27),

$$|\phi_M(u,T)| \le e^{C_T}\, e^{-\frac{1}{2}\sigma_T^2 u^2}, \quad u \in \mathbb{R}, \tag{24}$$

where $C_T := \sum_{j=1}^{2M} |A_j(T)|/|\mathrm{Im} u_j(T)| < \infty$.

Remark 20. [Pre-asymptotic regime] The bound (24) is governed by C_T, which blows up if any pole approaches $\mathbb{R}$. The Gaussian envelope dominates e^{C_T} for $|u| \gtrsim \sqrt{2\, C_T/\sigma_T^2}$. For smaller $|u|$ the trivial bound $|\phi_M(u,T)| \le 1 + \sum_j |A_j(T)|/|\mathrm{Im} u_j(T)|$ is used.

Remark 21. [Tail comparison] Padé ϕ_M has Gaussian decay $e^{-\frac{1}{2}\sigma_T^2 u^2}$, sharper than the stretched-exponential $e^{-c|u|^{2\mu}}$ $(\mu < 1)$ of the exact rough Heston cf.

Proposition 22. *[Strip admissibility] For rough Heston with* $\mu \in (\frac{1}{2}, 1)$ *and Padé order* M, *the PFE poles* $\{u_j(T)\}_{j=1}^{2M}$ *split by Hermitian symmetry (Lemma 18) into* M *poles in* $\{\mathrm{Im} u_j < 0\}$ *and* M *in* $\{\mathrm{Im} u_j > 0\}$. *Consider*

$$\min_{j=1,\ldots,2M} \mathrm{Im} u_j(T) > c + 1 \tag{25}$$

for $c \in \mathbb{R}$. *Any* $c \le \min_j \mathrm{Im} u_j(T) - 1$ *satisfies* (25); *such* c *exists because the PFE has finitely many poles off* $\mathbb{R}$. *The corresponding damping* $\alpha = -c - 1$ *is positive and finite.*

Remark 23. [Independence from c] By Cauchy's theorem the Lewis contour can be deformed between any two admissible vertical lines without crossing poles, so the integral value is unchanged. The Fox H price formula (34) is a contour-invariant analytic identity.

6 Closed-form density via Fox *H*

Definition 24. *[Fox* H*-function] For nonnegative integers* m, n, p, q *with* $0 \le n \le p$, $0 \le m \le q$,

$$H_{p,q}^{m,n}\left[z \,\middle|\, \begin{matrix} (a_i,\alpha_i)_1^p \\ (b_j,\beta_j)_1^q \end{matrix} \right] := \frac{1}{2\pi i} \int_{\mathcal{L}} \frac{\prod_{j=1}^m \Gamma(b_j + \beta_j s) \prod_{i=1}^n \Gamma(1 - a_i - \alpha_i s)}{\prod_{j=m+1}^q \Gamma(1 - b_j - \beta_j s) \prod_{i=n+1}^p \Gamma(a_i + \alpha_i s)}\, z^{-s}\, ds. \tag{26}$$

Theorem 25. *[Closed-form density] With PFE*

$$\Phi_M(u,T) = -\tfrac{1}{2}\sigma_T^2 u^2 - i\,\mu_T u + \sum_{j=1}^{2M} \frac{A_j(T)}{u - u_j(T)}, \tag{27}$$

$\sigma_T^2 > 0$, $\mu_T \in \mathbb{R}$, $u_j \notin \mathbb{R}$, $\epsilon_j := -\mathrm{sgn}(\mathrm{Im} u_j(T))$, $f_M(x,T) := \mathcal{F}^{-1}[\phi_M(\cdot,T)](x)$ *admits*

$$f_M(x,T) = \sum_{S \in \mathcal{P}(\{1,\ldots,2M\})} d_S(T)\; H^{m_S^\circ, n_S^\circ}_{p_S^\circ, q_S^\circ}\left[\eta_S(x,T) \,\middle|\, \begin{matrix} (\mathbf{a}_S^\circ, \boldsymbol{\alpha}_S^\circ) \\ (\mathbf{b}_S^\circ, \boldsymbol{\beta}_S^\circ) \end{matrix}\right] \tag{28}$$

with $(m_S^\circ, n_S^\circ, p_S^\circ, q_S^\circ) = (|S|+1, |S|, |S|+1, 2|S|+1)$,

$$d_S(T) = \frac{1}{\sqrt{2\,\pi\,\sigma_T^2}} \prod_{j \in S} \epsilon_j \sqrt{A_j(T)/i}, \tag{29}$$

$$\eta_S(x,T) = \frac{(x-\mu_T)\; 1_{\bigwedge_{j\in S}\epsilon_j (x-\mu_T) > 0}}{\sqrt{2\,\sigma_T^2}} \prod_{j\in S} (i\,A_j(T)) \cdot \exp\!\left(-i \sum_{j \in S} u_j(T)\,(x-\mu_T)\right), \tag{30}$$

$$\begin{aligned} \mathbf{a}_S^\circ &= (0,\ldots,0)_{|S|}, & \boldsymbol{\alpha}_S^\circ &= (1,\ldots,1)_{|S|}, \\ \mathbf{b}_S^\circ &= (0_{\text{Gauss}}, (0)_{|S|}, (-1)_{|S|}), & \boldsymbol{\beta}_S^\circ &= (\tfrac{1}{2}, (1)_{|S|}, (1)_{|S|}). \end{aligned} \tag{31}$$

All entries are constant in $j \in S$. *The* j*-dependent data enters only through* $d_S(T)$ *and* $\eta_S(x,T)$, *both symmetric in* S.

Corollary 26. *[Newton–Girard symmetric collapse] The level-*r *consolidations*

$$\sum_{|S|=r} \prod_{j \in S} (i\,A_j(T)) = i^r\; e_r\,(A_1(T), \ldots, A_{2M}(T)), \tag{32}$$

$$\sum_{|S|=r} \exp\!\left(-i \sum_{j\in S} u_j(T)\,\xi\right) = e_r\,(e^{-i u_1(T)\xi}, \ldots, e^{-i u_{2M}(T)\xi}), \tag{33}$$

reduce the 2^{2M}*-subset sum to* $2M+1$ *symmetric groups computable in* $O(M \log^2 M)$ *via FFT polynomial multiplication of* $\prod_j (1 + t\,x_j)$.

7 Closed-form European call price via Fox H

Theorem 27. *[Closed-form European call price] With* $\tilde{k} := \log(K/S_0) - rT$,

$$C_M(K,T) = S_0 \sum_{S \in \mathcal{P}(\{1,\ldots,2M\})} c_S(T)\ H^{m_S,n_S}_{p_S,q_S}\left[\zeta_S(K,T)\,\middle|\,\begin{matrix}(\mathbf{a}_S,\boldsymbol{\alpha}_S)\\(\mathbf{b}_S,\boldsymbol{\beta}_S)\end{matrix}\right] \tag{34}$$

with $(m_S,n_S,p_S,q_S) = (|S|+1, |S|+1, |S|+2, 2|S|+2)$,

$$c_S(T) = \frac{1}{\sqrt{2\pi\sigma_T^2}} \prod_{j\in S} \epsilon_j \sqrt{A_j(T)/i}, \tag{35}$$

$$\zeta_S(K,T) = \frac{(K/S_0)\,e^{-rT-\mu_T}}{\sqrt{2\sigma_T^2}} \prod_{j\in S} (i\,A_j(T)). \tag{36}$$

After Newton–Girard the sum reduces to $2M+1$ *symmetric groups, evaluable in* $O(M\log^2 M)$.

Theorem 25 gives (28). The payoff splits as $(S_T-K)^+ = S_0\,e^{x+rT}\,1_{x>\tilde{k}} - K\,1_{x>\tilde{k}}$. Per Fox H-summand,

$$\int_{\tilde{k}}^{\infty} H^{m,n}_{p,q}[\zeta x|\cdots]\,dx = \zeta^{-1} H^{m+1,n}_{p+1,q+1}[\zeta\tilde{k}|(0,1),\ldots,(1,1)], \tag{37}$$

adding one Γ-factor each in (m,p,q). The Mellin shift followed by (37) adds one in n. The increment gives $(|S|+1, |S|+1, |S|+2, 2|S|+2)$.

Assume (25) for some $c\in\mathbb{R}$ (Proposition 22). The Lewis representation

$$C_M(K,T) = \frac{S_0\,e^{-rT}}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{\phi_M(iw,T)\,e^{-w\tilde{k}}}{w(w-1)}\,dw \tag{38}$$

combined with the Bessel resummation

$$\sum_{n\geq 1} \frac{c^n}{n!(s-u)_n} = \frac{\Gamma(s-u)\,I_{s-u-1}(2\sqrt{c})}{c^{(s-u-1)/2}} - 1, \tag{39}$$

which converges absolutely on $\mathrm{Re}(s-u)>0$, identifies the result as Fox H with the same parameters.

Corollary 28. *[Two-route verification] (34) is established by two analytically independent routes converging on identical parameter data.*

Corollary 29. *[Quadrature-free pricing]* $C_M(K,T)$ *is a finite sum of Fox* H *evaluations, each via an absolutely convergent residue series with* n*-th term* $O(n!^{-1})$*, no Fourier inversion quadrature, no cosine series, no truncation interval.*

8 COS pricing as alternative

8.1 Cumulants

$\psi_M(s,T) := \Phi_M(-i\,s,T)$,

$$\kappa_n^{(M)}(T) := \frac{d^n\,\psi_M(s,T)}{d\,s^n}\Big|_{s=0}, \quad n=1,2,3,4. \tag{40}$$

8.2 Truncation interval

$$[a(T),b(T)] := \left[\kappa_1^{(M)} - L\sqrt{\kappa_2^{(M)} + \sqrt{\kappa_4^{(M)}}},\ \kappa_1^{(M)} + L\sqrt{\kappa_2^{(M)} + \sqrt{\kappa_4^{(M)}}}\right], \quad L \in [8,12]. \tag{41}$$

Remark 30. [Single-interval term structure] $\kappa_2^{(M)}(T), \kappa_4^{(M)}(T)$ are nondecreasing in T, so $[a,b] := [a(T_{\max}), b(T_{\max})]$ is admissible for every $T \in (0, T_{\max}]$.

8.3 Payoff coefficients

$w(x) = K\,(e^x - 1)^+$, $\tilde{U}_n = \frac{2}{b-a}\,[\chi_n(0,b) - \psi_n(0,b)]$,

$$\begin{aligned}\chi_n(c,d) = &\frac{1}{1+(n\,\pi\,/\,(b-a))^2}\,\Big[\cos\,(n\,\pi\,(d-a)\,/\,(b-a))\,e^d - \cos\,(n\,\pi\,(c-a)\,/\,(b-a))\,e^c \\ &+ \tfrac{n\,\pi}{b-a}(\sin\,(n\,\pi\,(d-a)\,/\,(b-a))\,e^d - \sin\,(n\,\pi\,(c-a)\,/\,(b-a))\,e^c)\Big],\end{aligned} \tag{42}$$

$$\psi_n(c,d) = \begin{cases} d-c, & n=0, \\ \frac{b-a}{n\,\pi}\,[\sin\,(n\,\pi\,(d-a)\,/\,(b-a)) - \sin\,(n\,\pi\,(c-a)\,/\,(b-a))], & n \geq 1. \end{cases} \tag{43}$$

8.4 COS price with Gaussian-rate error

Theorem 31. *[COS pricing]* $x_0 := \log\,(S_0\,/\,K) + r\,T$,

$$C_{M,N} := K\,e^{-rT} \sum_{n=0}^{N-1}{}' \operatorname{Re}\Big[\phi_M\Big(\frac{n\,\pi}{b-a}, T\Big)\exp\Big(i\,n\,\pi\frac{x_0 - a}{b-a}\Big)\Big]\,\tilde{U}_n. \tag{44}$$

$C_{M,N} \to C_M$ *as* $N \to \infty$, *with*

$$|C_M - C_{M,N}| \leq C'\exp\Big(-\tfrac{1}{2}\sigma_T^2\,(N\,\pi\,/\,(b-a))^2\Big). \tag{45}$$

9 Algorithm and computational summary

The Fox H pricing endpoint (34) consumes the PFE data $\{u_j(T), A_j(T), \sigma_T, \mu_T\}$ and accepts K directly through $\zeta_S\,(K,T)$; no Fourier inversion is performed and no grid in v is required.

9.1 Three-tier decomposition

Tier 0 (per parameter evaluation, per Fourier node). With v specialised to a numeric value, the Chebyshev algorithm (Theorem 8) on $\{M_l(v)\}_{l=0}^{2M-1}$ costs $O(M^2)$ scalar operations.

Tier 1 (per maturity T). Specialise $z=T^\mu$ to obtain the rational $\Phi_M(v,T)$, then compute the PFE $\{u_j(T), A_j(T), \sigma_T, \mu_T\}_{j=1}^{2M}$ by root-finding and cover-up at $O(M^2)$ per T.

Tier 2 (per strike K at fixed T). Fox H evaluation (34) via Newton–Girard consolidation (Corollary 26) and an absolutely convergent residue series with $n!^{-1}$ terms, $O(M\log^2 M)$ per (K,T), no quadrature.

9.2 Total cost across an (N_T, N_K)-surface

For one parameter evaluation,

$$\underbrace{O(M^2)}_{\text{Tier 0}} + \underbrace{N_T\cdot O(M^2)}_{\text{Tier 1, per } T} + \underbrace{N_T N_K\cdot O(M\log^2 M)}_{\text{Tier 2, per } (K,T)}. \tag{46}$$

The structurally novel feature is the K-independence at Tier 2: every strike at fixed T shares the PFE data and costs only $O(M\log^2 M)$.

9.3 COS endpoint

The COS endpoint requires $\phi_M(v_n,T)$ on a Fourier grid, with each node costing $O(M)$ via the rational form of Φ_M, giving $O(NM)$ per (K,T).

Remark 32. [Cost decomposition] Per parameter evaluation, the pipeline scales as $O(M^2)$ for the Chebyshev front-end, $O(M^2)$ per maturity for the PFE, and $O(M\log^2 M)$ per (K,T) via Fox H, with no quadrature, no Fourier inversion, and no truncation interval. The K-independence of Tier 2 is what allows the full strike slice at fixed T to be priced in time proportional to the slice size, not to its product with M.

10 Summary

T1. Padé resummation directly on $\{M_k(v)\}$ via the Chebyshev algorithm in arbitrary precision, in $O(M^2)$ ring operations with no Hankel matrix inversion. Stahl convergence (Theorem 14) extends beyond the Puiseux radius. The supremum-ratio a-posteriori bound (Theorem 15) fixes M from precision.

T2. Density (Theorem 25) and call price (Theorem 27) admit strictly closed-form Fox H representations, evaluable in $O(M\log^2 M)$ per (K,T) with no quadrature; the front-end is $O(M^2)$ per parameter evaluation and the PFE $O(M^2)$ per maturity (Remark 32). Established by two analytically independent routes (Corollary 28).

T3. COS pricing (Theorem 31) on rational $\Phi_M(\cdot,T)$ provides a fallback endpoint of cost $O(NM)$ per (K,T) with Gaussian-rate truncation error.

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